

Tongren Xiao

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Research Interests

Reliable language agents and structured decision support; uncertainty-aware forecasting and probabilistic world models; tool use and auditable AI systems for financial and scientific applications.

Education

Sun Yat-sen University, School of Mathematics

2023–September 2027 (expected)

Undergraduate Student in Mathematics

GPA: 3.00/5.00; average mark: 80/100

Research Experience

Research in Quantitative Finance

2024–2026

Advisor: Associate Professor Yirong Huang, Sun Yat-sen University

- Studied hybrid LSTM–GARCH, geometric analog forecasting, forecast calibration, and underestimation-risk control for realized volatility.
- Designed comparative experiments and risk-sensitive diagnostics for financial time-series forecasting.

Undergraduate Research in Schubert Calculus

2024–2026

Advisor: Professor Changzheng Li, Sun Yat-sen University

- Studied positivity properties of quantum double Schubert polynomials and related algebraic-combinatorial structures.

Independent Mathematical Research

2026–present

Research on matrix-analysis and Diophantine conjectures, with emphasis on proof-oriented problem formulation.

Selected Research Projects

SRM Forecasting Agent

2026

- Developed an auditable forecasting-agent interface supporting natural-language requests, data-adaptor selection, information-boundary validation, geometric retrieval, and JSON audit artifacts.
- Added an optional cloud LLM tool-use prototype with schema validation and grounded explanations; the five-channel SRM engine and paper protocol remain unchanged.
- Code and reproducibility documentation

Probabilistic World Model for Multi-Asset Volatility

2026

- Built a recurrent state-space model with shared market dynamics and asset-specific latent representations for probabilistic realized-volatility paths.
- Enforced a forward-only information boundary and evaluated joint path realism with Monte Carlo forecasts and reproducible audit manifests.
- Code and paper

Selected Manuscripts and Outputs

- *Shape Is Useful: Geometric Analog Forecasting of Realized Volatility*. Manuscript, 2026.
- *A Comparative Study of Hybrid LSTM Frameworks for Volatility Forecasting*. Manuscript, 2025.
- *Graham Positivity of Quantum Double Schubert Polynomials*. Manuscript, 2026.
- *Anchor-Conditioned Probabilistic World Models for Multi-Asset Realized-Volatility Paths*. Work in progress, 2026.

Technical Skills

Python, MATLAB, NumPy, pandas, PyTorch, scikit-learn, Streamlit, FastAPI, Git, Linux, LaTeX; probabilistic modeling, time-series forecasting, reproducible experiments, structured tool interfaces.

Awards and Training

Honorable Mention, 2026 COMAP Mathematical Contest in Modeling; Guangdong Provincial Second Prize, 2025 Contemporary Undergraduate Mathematical Contest in Modeling; Introduction to Option Pricing (Johannes Ruf), grade 81.50.